

Inferring Submesoscale Surface Velocity and Vertical Velocity from SWOT Surface Sea Height Data

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Motivation

Submesoscale acts as the bridge in the energy cascade, moving energy from the geostrophic mesoscales down to the small scales where dissipation occurs. The **SWOT** resolves sea surface height (SSH) down to $\mathcal{O}(10\text{km})$. We wish to develop a method to infer submesoscale surface velocity and vertical velocity from **SWOT** surface sea height data.

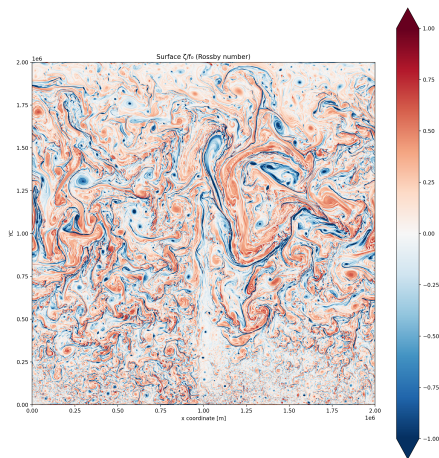


Figure: A Surface Vorticity field from MITgcm simulation of the Southern Ocean.

Interior PV equation (zero-PV)

$$\nabla_h^2 \psi + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) = 0 \quad (1)$$

Surface boundary condition

$$f_0 \left. \frac{\partial \psi}{\partial z} \right|_{z=0} = b_s, \quad \psi \rightarrow 0 \text{ as } z \rightarrow -\infty \quad (2)$$

- SSH gives surface buoyancy via geostrophic balance: $b_s \propto g \eta$.
- The full interior flow $(\psi, \mathbf{u}_g)$ is determined by b_s alone.

In 2022, a new satellite called SWOT was launched by NASA, CNES to access the mesoscale and submesoscale dynamics of the ocean. SSH will be measured by the two **KaRin** (Ka-band Radar Interferometer) instruments. They will provide a 120 km width swath of coverage with a resolution of 2 km.

Methodology — The SQG⁺¹ Model

[Muraki et al., 1999]

Limitation of SQG

SQG is $\mathcal{O}(\text{Ro}^0)$: captures geostrophic balance but **misses** ageostrophic corrections responsible for frontogenesis and vertical motion.

SQG⁺¹ extension

Expand in Rossby number,

$$\psi = \underbrace{\psi^{(0)}}_{\text{SQG}} + \text{Ro} \psi^{(1)} + \dots$$

The $\mathcal{O}(\text{Ro})$ correction is driven nonlinearly by the SQG base state.

Key advantage

Yields a non-trivial vertical velocity w and ageostrophic surface velocity — all from SSH alone.

SQG⁺ vorticity decomposition

$$\zeta = \underbrace{\nabla^2 \Phi^0}_{\zeta^0, \text{SQG}} + \text{Ro} \underbrace{(\nabla^2 \Phi^1 + F_{yz}^1 - G_{xz}^1)}_{\zeta^1, \text{correction}} \quad (3)$$

- Φ^0 : SQG streamfunction, determined by $\Phi^{0,s}$ with exponential vertical decay in spectral space.
- Φ^1, F^1, G^1 : all nonlinear functions of $\Phi^{0,s}$ — no new inputs needed.

Cyclogeostrophic pressure correction

$$\nabla^2 p^1 - f \zeta^1 = 2J(\Phi_x^{0,s}, \Phi_y^{0,s}) \quad (4)$$

The Jacobian on the right captures the **ageostrophic** feedback from the SQG base state.

Methodology — Forward Map II: SSH

First-order SSH

From Eq. (4), under hydrostatic balance,

$$\eta^1 = \mathcal{N}(\Phi^{0,s}) \quad (5)$$

where $\mathcal{N}$ is a nonlinear operator.

Full model SSH

$$\eta_{\text{model}} = \underbrace{\Phi^{0,s}}_{\text{SQG}} + \epsilon \underbrace{\mathcal{N}(\Phi^{0,s})}_{\text{correction}} \quad (6)$$

η_{model} depends *only* on the surface streamfunction $\Phi^{0,s}$ — this is the key to the inversion.

Methodology — The Inverse Problem

Control variable

Everything in SQG^{+1} is determined by $\Phi^{0,s}$. **Goal:** find $\Phi^{0,s}$ such that model SSH matches observations.

Cost function

$$\min_{\Phi^{0,s}} \|\eta_{\text{obs}} - \eta_{\text{model}}(\Phi^{0,s})\|_2^2 \quad (7)$$

The optimization is done by using a limited-memory L-BFGS algorithm. The gradient is computed automatically through the python JAX library.

About the SWOT Data — Potential Challenges

Since the ultimate goal is to apply this method to SWOT observations, there are several challenges to address.

Challenges

- **SWOT observation errors:** The SSH retrieved from SWOT is not a perfect signal. The Ka-band Radar Interferometer is highly sensitive to random *thermal noise* in the electronics, and other error sources (KaRIn roll, wet-troposphere delay, etc.) further contaminate the small-scale SSH field that SQG^{+1} is most sensitive to. [Chelton et al., 2022]
- **Non-steady signals:** SQG^{+1} assumes a steady-state flow, but SWOT observes tides, internal waves, and transient eddies. These unsteady features cannot be captured by the SQG^{+1} model and need to be filtered out.

Objective

Test the SQG^{+1} inversion on a suite of *synthetic* perturbations, each designed to mimic a specific class of ageostrophic signal.

Running the inversion directly on SWOT data — or even on a primitive-equation channel simulation — conflates every source of non-SQG signal at once. Because the individual contributions cannot be disentangled, it is impossible to attribute errors to a particular mechanism or to design targeted, controlled experiments. The final goal is to improve the current reversion method given conclusions drawn from the benchmark results.

Data and Target Field

- Use an SQG simulation from Prof. Smith as the balanced base field $\Phi_{\text{SQG}}^{0,s}$.
- Apply the SQG^{+1} forward operator to obtain a clean target SSH, then add an $\mathcal{O}(\text{Ro})$ perturbation: $\eta^{\text{true}} = \mathcal{N}^{\text{tot}}(\Phi_{\text{SQG}}^{0,s}) + p$.
- Each perturbation p is normalized so that $\text{rms}(p) = \text{Ro} \cdot \text{rms}(\Phi_{\text{SQG}}^{0,s})$ — the initial cost is equal across cases, so the benchmark isolates the *structure* of the signal rather than its amplitude.
- Run the SQG^{+1} inversion on each case and compare the recovered field against $\Phi_{\text{SQG}}^{0,s}$.

Below I list the perturbation families of interest. I am still working on testing

Group A — SQG-correlated structures

- **Frontogenesis:** $p \propto |\nabla\psi|^2$, amplified at fronts and strain regions.
- **Vorticity-aligned:** $p \propto \zeta = -\nabla^2\psi$ amplified inside vortex cores.

Group B — Inertia-gravity waves

- **Monochromatic IGW:** a single plane wave $\cos(k_0(\cos\theta x + \sin\theta y))$ pointing a given position.
- **IGW packet:** superposition of 8 plane waves spanning from meso- to submesoscale ($k = 5\text{--}30$) with random phase.

Group C — Spectrally shaped white noise

- **High- k block:** white noise restricted to $k > k_c$, where k_c is some cutoff threshold. Mimicking instrument and submesoscale contamination.
- **Low- k block:** white noise restricted to $k < k_c$, testing sensitivity at the large-scale end of the spectrum.
- **K^{-n} spectrum:** noise reshaped so its energy decays as K^{-n} , where n is some exponent. Mimicking the observed slope of the oceanic SSH spectrum.

Group E — SWOT-like errors

- **KaRIn-like noise:** spectrally anisotropic noise that decorrelates faster in the cross-track direction than along-track — the signature of KaRIn baseline error [Chelton et al., 2022, Gaultier et al., 2016].
- **Roll error:** slowly-varying along-track amplitude modulating a domain-scale cross-track pattern, the spectral fingerprint of residual antenna roll [Gaultier et al., 2016, Esteban-Fernández, 2017].

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